FINAL REPORT

Test Facility Study No. 9601568

SM-102
In Vitro Mammalian Cell Micronucleus Test in Human Peripheral Blood Lymphocytes

SPONSOR:
Moderna Therapeutics, Inc.
200 Technology Square, Third Floor
Cambridge, MA 02139
USA

TEST FACILITY:
Charles River Laboratories Montreal ULC
(b) (6)
[redacted]

12 January 2017

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COMPLIANCE STATEMENT

The study was performed in accordance with the OECD Principles of Good Laboratory Practice and as accepted by Regulatory Authorities throughout the European Union, United States of America (FDA), Japan (MHLW), and other countries that are signatories to the OECD Mutual Acceptance of Data Agreement.

Exceptions from the above regulations are listed below.

- Characterization of the Test Item was performed by the Sponsor subcontractor according to established SOPs, controls and approved test methodologies to ensure integrity and validity of the results generated; these analyses will not be conducted in compliance with the GLP or GMP regulations.
- Stability testing of the supplied Test Item was not determined in this study. It will be performed by the Sponsor subcontractor at a laboratory that follows FDA Good Manufacturing Practice (GMP) regulations.

This study was conducted in accordance with the procedures described herein. All deviations authorized/acknowledged by the Study Director are documented in the Study Records. The report represents an accurate and complete record of the results obtained.

There were no deviations from the above regulations that affected the overall integrity of the study or the interpretation of the study results and conclusions.

(b) (6) ________________ Date: 12 Jan 2017

QUALITY ASSURANCE STATEMENT

Study Number: 9601568

This Study has been audited by Quality Assurance in accordance with the applicable Good Laboratory Practice regulations. Reports were submitted in accordance with SOPs as follows:

QA INSPECTION DATES

		Dates Findings Submitted to:	
Date(s) of Audit	Phase(s) Audited	Study Director	Study Director Management
12-Sep-2016	Final Study Plan	12-Sep-2016	12-Sep-2016
12-Sep-2016	Dose Preparation	12-Sep-2016	12-Sep-2016
18-Nov-2016 21-Nov-2016 - 22-Nov-2016	Data Review - In Vitro Sciences	07-Dec-2016	07-Dec-2016
21-Nov-2016 - 22-Nov-2016	Data Review - Analytical Chemistry	23-Nov-2016	23-Nov-2016
21-Nov-2016 - 22-Nov-2016	Final Report	07-Dec-2016	07-Dec-2016
22-Nov-2016	Final Phase Report - Dose Formulation Analysis	23-Nov-2016	23-Nov-2016
09-Jan-2017	Study Plan Amendment 1	09-Jan-2017	09-Jan-2017

In addition to the above-mentioned audits, process-based and/or routine facility inspections were also conducted during the course of this study. Inspection findings, if any, specific to this study were reported by Quality Assurance to the Study Director and Management and listed as a Phase Audit on this Quality Assurance Statement.

The Final Report has been reviewed to assure that it accurately describes the materials and methods, and that the reported results accurately reflect the raw data.

(b) (6)

12-Jan-2017
Date

1. RESPONSIBLE PERSONNEL

1.1. Test Facility

Study Director	(b) (6)
Test Facility Management	(b) (6)

1.2. Individual Scientists (IS) at Test Facility

Dose Formulation Analysis	(b) (6)

2. SUMMARY

The objective of this study was to determine the potential genotoxicity of SM-102, using an in vitro mammalian cell micronucleus test in human peripheral blood lymphocytes.

The experimental design was as follows:

Text Table 1
Main Test

Dose No.	Formulation Conc. (µg/mL)[a]	Final Conc. (µg/mL)	Number of Cultures		
			4 Hours (0S9)	4 Hours (+S9)	24 Hours (0S9)
Negative Control	-	-	2	2	2
1/ SM-102	325	3.25	2	2	2
2/ SM-102	568	5.68	2	2	2
3/ SM-102	995	9.95	2	2	2
4/ SM-102	1740	17.4	2	2	2
5/ SM-102	3050	30.5	2	2	2
6/ SM-102	5330	53.3	2	2	2
7/ SM-102	9330	93.3	2	2	2
8/ SM-102	16300	163	2	2	2
9/ SM-102	28600	286	2	2	2
10/ SM-102	50000	500	2	2	2
NOC	25	0.25	2	-	-
	30	0.30	2	-	-
CP	1000	10	-	2	-
	1500	15	-	2	-
MMC	10	0.10	-	-	2
	20	0.20	-	-	2

a Theoretical concentrations; actual concentrations may differ slightly due to the limitations of the instruments used.
b Where the high level = 0.5 mg/mL

Human peripheral blood lymphocytes were treated in the absence and presence of an Aroclor-induced S9 activation system for 4 hours and continuously for 24 hours. The positive controls caused statistically significant increases in the incidence of micronucleated binucleate cells in each regime of the study, confirming the sensitivity of the test system and the effectiveness of the S9 mix.

Cultures treated with SM-102 at levels up to 500 µg/mL, did not show any statistically significant increases in the incidence of micronucleated binucleate cells. Precipitation was observed at the end of treatment at the highest dose level tested, in the 4 hour regime and the 24 hour regime in the absence of S9 mix. No precipitate was observed in the presence of S9 mix. Cloudy media was observed in the 4 hour regime in the absence of S9 mix at dose levels ≥ 93.3 µg/mL, in the 4 hour regime in the presence of S9 mix at the highest dose level and in the 24 hour regime at dose levels ≥ 286 µg/mL. No cytotoxicity was observed in the assay.

It is concluded that SM-102 did not show any evidence of genotoxic activity in this in vitro test for induction of micronuclei in human peripheral blood lymphocytes, when tested in accordance with regulatory guidelines.

3. INTRODUCTION

The objective of this study was to determine the potential genotoxicity of SM-102, using an in vitro mammalian cell micronucleus test in human peripheral blood lymphocytes.

The design of this study was based on OECD Guideline 487 and ICH Guideline S2(R1).

The Study Director signed the Study Plan on 08 Sep 2016, and dosing was initiated on 15 Sep 2016. The experimental start date was 12 Sep 2016, and the experimental completion date was 26 Oct 2016. The study was completed on the date of the Study Director approval of this report (refer to the appropriate signature page). The Study Plan, last Study Plan amendment, and deviations are presented in Appendix 1.

4. MATERIALS AND METHODS

4.1. Test and Reference Items

4.1.1. Test Item

Identification:	SM-102
Batch (Lot) No.:	RL-100-211-1
Retest Date:	27 Oct 2017
Molecular Weight:	710.18
Purity:	95.72% (All concentrations and dose levels throughout this report were corrected for purity using a purity of 95.3%.)
Storage Conditions:	Kept in a freezer set to maintain -20°C

4.1.2. Reference Items

4.1.2.1. Negative Control

Identification:	Ethanol
Supplier:	Commercial Alcohols
Batch /Lot No.:	020612
Expiration Date:	Jun 2017
Storage Conditions:	Kept at ambient room temperature

4.1.2.2. Positive Controls

In the Absence of S9 Mix

Identification:	Mitomycin C (MMC)
CAS No.:	50-07-7
Identity:	Nocodazole (NOC)
CAS No.:	31430-18-9

In the Presence of S9 Mix

Identification: Cyclophosphamide monohydrate (cyclophosphamide, CP)

CAS No.: 6055-19-2

Full details of positive controls including supplier, storage, expiry date and formulation are retained as part of the Test Facility records (MMC, NOC) or study records (CP). Copies of certificates of analysis are retained as raw data.

4.2. Test Item Characterization

The Sponsor provided to the Test Facility documentation of the identity, strength, purity, composition, and stability for the Test Item. A Certificate of Analysis was provided to the Test Facility and is presented in Appendix 2.

4.3. Analysis of Test Item

A Certificate of Analysis was provided by the Sponsor and is presented in Appendix 2.

4.4. Test Item Inventory and Disposition

Records of the receipt, distribution, and storage of Test Item were maintained with the study raw data. All unused Test Item was returned to the Sponsor following completion of the experimental phase of the study. Any remaining Reference Items (negative and positive controls) will be retained at the Test Facility or discarded upon expiry.

4.5. Dose Formulation and Analysis

4.5.1. Preparation of Reference Items

4.5.1.1. Preparation of Negative Control

An adequate amount of the Negative Control, ethanol, was dispensed into a vial for administration to control cultures. The aliquot was stored in a refrigerator set to maintain 4°C until use. Any residual volumes were discarded before issuance of the Final Report.

4.5.1.2. Preparation of Positive Controls

Mitomycin C and Nocodazole were prepared the day of use or up to 6 months prior to use; adequate amounts were dispensed into vials, and stored in a freezer set to maintain -80°C (MMC) or -20°C (NOC), protected from light, until use. The aliquots were removed from the freezer and allowed to warm to ambient room temperature before dosing. Cyclophosphamide was prepared on the day of use. Any residual volumes were discarded after completion of dosing.

4.5.2. Preparation of Test Item

The Test Item was prepared as a stock solution (50 mg/mL) in the chosen vehicle (ethanol) and all lower level formulations were made by serial dilution. The formulations were prepared 3 days prior to use and were stored in a refrigerator set to maintain 4°C until use. The formulations were removed from the refrigerator and allowed to warm to room temperature for at least 30 minutes before dosing. Any residual volumes were discarded before issuance of the Final Report.

4.5.3. Sample Collection and Analysis

Samples from dose formulations 1 to 10 were collected for concentration analysis only. Homogeneity, density and stability were not determined on these samples. The positive control formulations were not subjected to analysis for safety reasons and because the biological response of the test system is considered to be the best measure of the appropriateness of the formulations.

Samples to be analyzed were transferred at ambient room temperature, to the Analytical Chemistry department at the Test Facility on the date prepared. Any residual/retained analytical samples were discarded before issuance of the Final Report.

4.5.3.1. Analytical Method

Analyses were performed by HPLC, using a validated analytical procedure (Test Facility Study Number 1801841).

4.5.3.2. Concentration Analysis

Duplicate 1 mL samples for dose numbers 1 and 2 and duplicate 0.5 mL samples for dose numbers 3 to 10 were taken and sent to the analytical laboratory for analysis. Additional duplicate 1 mL samples for dose numbers 1 and 2 and duplicate 0.5 mL samples for dose numbers 3 to 10 were taken and retained at the Test Facility as backup samples. Concentration results were considered acceptable if sample concentration results were within ±10% of nominal for the stock solution and ±15% of nominal for lower level solutions. After acceptance of the analytical results, backup samples were discarded.

4.5.3.3. Stability Analysis

Stability analyses performed previously at the Test Facility under Study Number 1801841 demonstrated that the Test Item is stable in the vehicle when prepared and stored under the same conditions at concentrations bracketing those used in the present study. Stability data have been retained in the study records for Study No. 1801841.

4.6. Test System

Primary cultures of human peripheral lymphocytes are recommended because of their low and stable background frequency of micronucleus formation. In addition, human cells are generally the most relevant for risk assessment.

4.6.1. Justification for Dose Level Selection

Typically, the Test Item is dosed at a range of concentrations, but is only assessed at the highest levels below the toxic level or, if non-toxic, at levels up to the standard limit of 0.5 mg/mL or 1 mM, whichever is lower (ICH S2(R1) guideline). Compounds with limited aqueous solubility are tested up to a level expected to show visible precipitation in the culture medium at the end of treatment.

4.6.2. Blood Sampling

A peripheral blood sample was taken by venipuncture from two young (approximately 18-35 years of age), healthy, non-smoking, male donors with no known recent exposures to genotoxic chemicals or radiation. The blood samples were collected directly into tubes containing sodium heparin and then held at room temperature for less than 2 hours prior to blood addition to the culture medium. Blood from both donors was pooled in the culture medium prior to culture initiation.

4.6.3. Culture Medium

Complete RPMI 1640 medium was prepared by supplementing RPMI 1640 medium with the following filter-sterilized components: 10% (v/v) fetal calf serum, 50 µg gentamycin per mL, and 4 units heparin per mL.

4.6.4. Lymphocyte Culture

Whole blood was mixed with medium (0.4 mL blood per 4 mL medium) and phytohemagglutinin (1 mL per 49 mL diluted blood) was added to stimulate lymphocyte division. Aliquots of 5 mL of cell suspension were dispensed into flat-sided culture tubes and then placed in an incubator set to maintain 37°C with 5% CO_2 in a humidified atmosphere.

4.7. S9 Mix

The S9 mix, used as a model of intact mammalian metabolism, was prepared on the day of use and contained 10% v/v S9 fraction (Aroclor 1254 induced male rat liver fraction supplied by Moltox) and the following sterile cofactors: 8 mM $MgCl_2$, 33 mM KCl, 100 mM sodium phosphate buffer pH 7.4, 5 mM glucose-6-phosphate, and 4 mM NADP. The S9 mix was stored in a refrigerator set to maintain 4°C or on ice until required. A copy of the manufacturer's quality control certificate for the S9 fraction is retained as raw data.

4.8. Experimental Design

Text Table 2
Main Test

Dose No.	Formulation Conc. (µg/mL)[a]	Final Conc. (µg/mL)	Number of Cultures		
			4 Hours (0S9)	4 Hours (+S9)	24 Hours (0S9)
Negative Control	-	-	2	2	2
1/ SM-102	325	3.25	2	2	2
2/ SM-102	568	5.68	2	2	2
3/ SM-102	995	9.95	2	2	2
4/ SM-102	1740	17.4	2	2	2
5/ SM-102	3050	30.5	2	2	2
6/ SM-102	5330	53.3	2	2	2
7/ SM-102	9330	93.3	2	2	2
8/ SM-102	16300	163	2	2	2
9/ SM-102	28600	286	2	2	2
10/ SM-102	50000	500	2	2	2
NOC	25	0.25	2	-	-
	30	0.30	2	-	-
CP	1000	10	-	2	-
	1500	15	-	2	-
MMC	10	0.10	-	-	2
	20	0.20	-	-	2

a = Theoretical concentrations; actual concentrations may differ slightly due to the limitations of the instruments used. b = Where the high level = 0.5 mg/mL.

4.9. Treatment

Treatments were performed approximately 48 hours (44-48 hours) after culture initiation. Appropriate dilutions of the Test Item and Positive Control formulations were prepared so as to reach the final concentrations indicated in the experimental design (see Text Table 2). Cultures tested in the absence of S9 mix were treated as indicated in the experimental design then returned to the incubator for 4 or 24 hours as appropriate. For cultures tested in the presence of S9 mix, 1 mL of S9 mix was added immediately prior to treatment, then the cultures were returned to the incubator for 4 hours. A standard dose volume of 10 µL Test Item, Negative Control or Positive Control per mL of culture was used throughout.

The Test Item was tested over a wide range of dose levels (3.25 to 500 µg/mL) using all treatment regimes (4-hour treatment period in the absence and presence of S9 mix and a 24-hour treatment period in the absence of S9 mix) so that analyzable cells would be available for at least three dose levels for each regime. Duplicate cultures were treated at each experimental point.

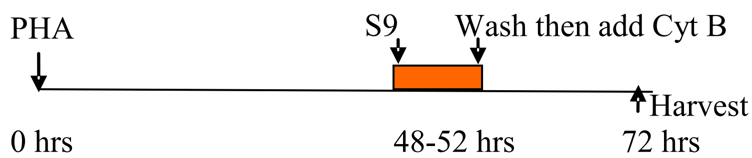
4.10. Medium Change (Wash) and Cytochalasin B (Cyt B) Treatment

After the 4-hour treatments, cultures were centrifuged, and the supernatant replaced with fresh complete medium containing 6 µg Cytochalasin B per mL. Incubation was continued for a further 20 hours prior to harvesting.

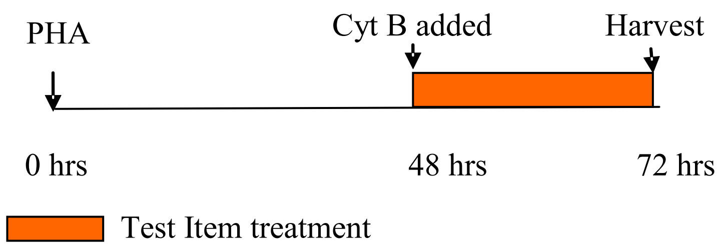
For the 24-hour treatment, the medium added to the cultures just prior to dosing contained 6 µg Cytochalasin B per mL of culture medium. The 24-hour treatment cultures were harvested at the end of the treatment period.

Text Figure 1

4-hour treatment in the absence or presence of S9 mix (+S9)



24-hour treatment in the absence of S9 (0S9)



4.11. Harvesting and Staining

Cultures were harvested by centrifugation. The cell pellet was resuspended in hypotonic solution (0.075 M KCl) at ambient temperature. Fixative (24:1 v/v, methanol:acetic acid) was mixed with the suspended cells and, following another centrifugation, the cells were treated with 3 changes of fixative. After the third change of fixative, the cell pellet was resuspended in fixative at an appropriate density for slide preparation. The fixed cells were dropped onto clean slides and air-dried before staining. Two or three slides were prepared from each culture. Fixed cells not used for slide preparation were discarded after completion of the experimental phase of study. Cells were stained with the fluorescent metachromatic dye, acridine orange. Stained fixed slides were kept for potential retrospective examination until study finalization.

4.12. Slide Examination

4.12.1. Cytokinesis-Block Proliferation Index (CBPI)

Slides were examined visually for toxicity and slides from a minimum of 3 concentrations up to the toxic concentration, or the lowest concentration which results in a visible precipitate in the culture medium at the end of treatment, whichever is least (or top concentration if there is no cytotoxicity or precipitate limitations) was selected for CBPI determination. Selected slides were randomized using Microsoft® Excel then encoded to minimize potential operator bias.

The CBPI was determined by examination of at least 500 cells (if available) per culture. Lymphocyte toxicity is normally indicated by a decreased CBPI compared to the concurrent negative control group.

$$\text{CBPI} = \frac{((\text{No. mononucleate cells}) + (2 \times \text{No. binucleate cells}) + (3 \times \text{No. multinucleate cells}))}{\text{Total number of cells}}$$

% Cytotoxicity = $100-100((CBPI_T-1) \div (CBPI_C-1))$

T = Test Item treatment group

C = Negative Control group

4.12.2. Selection of Slides for Micronucleus Assessment

Justification for selection of dose levels for examination is presented in the results section of the report. Routinely, slides for examination for micronuclei are selected primarily on the basis of the CBPI results. Normally, the highest level examined is the lowest concentration which results in approximately 55% toxicity, based on CBPI, with a sufficient number of scorable binucleate cells or the lowest concentration which causes precipitation at the end of treatment, whichever is least; in the absence of toxicity or precipitation, it is the highest dose tested. In addition, two lower dose levels, the Negative Control, and one dose level of the Positive Control for each regime are included in the detailed analysis.

4.12.3. Microscopic Analysis of Micronucleus Frequencies

Slides selected for analysis of micronucleus frequencies were examined by fluorescence microscopy using a blue excitation filter and a yellow barrier filter, and (where practical) a total of 2000 binucleate cells per experimental point (1000 per culture) were examined for the presence of micronuclei using oil-immersion optics.

Readable binucleate cells are identified by the following criteria:

- The cell must have two main nuclei.
- The two main nuclei must each have an intact and well-defined membrane.
- The two main nuclei must be contained within the cytoplasm.
- The cell must be visible in its entirety in the field.
- The area around the cell must not contain micronucleus-like debris.
- The cytoplasmic boundary should be intact and distinguishable from the boundaries of adjacent cells.

Micronuclei (MN) are identified by the following criteria:

- The diameter of the MN must not exceed 1/3rd of each of the two main nuclei diameter.
- The micronuclei can touch but must not overlap the two main nuclei.
- Micronuclei should be large enough to discern morphological characteristics.
- Micronuclei should possess a generally rounded shape with a clearly defined outline.

- Micronuclei should be similar in color to the nuclei.
- Should lie in the same focal plane as the cell.
- Micronuclei must not be linked to the nuclei by a nucleoplasmic bridge.
- Micronuclei must be within cytoplasmic boundary.
- Micronuclei must be non-refractive (staining).

The location (Vernier readings) of the first nine micronucleated binucleate cells (MBC) was recorded for potential peer review. No peer-review was required for this study.

The % micronucleated binucleate cells (% MBC) is the proportion of micronucleated binucleate cells over the total number of binucleate cells evaluated. Slides from cultures not subjected to analysis were discarded after completion of the experimental phase of study.

5. COMPUTERIZED SYSTEMS

Critical computerized systems used in the study are listed below or presented in the appropriate Phase Report. All computerized systems used in the conduct of this study have been validated; when a particular system has not satisfied all requirements, appropriate administrative and procedural controls were implemented to assure the quality and integrity of data.

Text Table 3
Critical Computerized Systems

System Name	Version No.	Description of Data Collected and/or Analyzed
Microsoft® Excel	2007	Slide randomization and arithmetic calculations of mean values, etc, for presentation in reports.
SAS	9.2	Statistical analyses of number of micronucleated cells
Provantis	8	Statistical analyses of the number of micronucleated cells where less than 2000 binucleated cells were examined per experimental point
Mesa Laboratories AmegaView CMS	v3.0 Build 1208.8	Continuous Monitoring System. Monitoring of standalone fridges, freezers, incubators, and selected laboratories to measure temperature, relative humidity, and CO_2, as appropriate
Johnson Controls Metasys	MVE 5.4 (M5)	Building Automation System. Control of HVAC and other building systems, as well as temperature/humidity control and trending in selected laboratories and animal rooms
Empower 3 (Waters Corporation)	Build 3471 SR1	Dose formulation analyses using HPLC.
Empower 3 and Microsoft Excel	Build 3471 SR1/ 2007	Regression analysis and descriptive statistics for dose formulation analytical data.

6. EVALUATION AND INTERPRETATION OF RESULTS

6.1. Assay Acceptance Criteria:

- *Acceptable Negative Control:* The incidence of micronucleated binucleate cells in the Negative Control should be considered acceptable for addition to the negative control data base (should be within the 95% control limits of the distribution of the Negative Control database or, where the incidence of micronucleated binucleate cells falls outside of the limits, should not be an extreme outlier and there is evidence that the test system is under control).
- *Acceptable Positive Control:* The number of micronucleated binucleate cells in the Positive Control cultures should be compatible with those generated in the historical Positive Control database and produce a statistically significant increase compared to the concurrent Negative Control.
- *Acceptable Cell Proliferation*: Cell proliferation, as measured by the CPBI, should indicate that the treatments are conducted at appropriate levels of cytotoxicity.
- *Experimental Conditions:* All three treatment regimes are to be used in the assay unless a positive result is obtained in one of the regimes.
- *Acceptable Number of Cells and Analyzable Concentrations:* An acceptable number of cells (see section 15) and at least three test concentrations are obtained.
- *Selection of Top Concentration:* The criteria for the selection of the top concentration are consistent with section 15.2.

In the event that the controls fall slightly outside the normal range (historical or Study Plan), the Study Director will be allowed discretion in accepting the results of the experiment as valid based on the biological significance.

6.2. Statistical Analysis

The results obtained for each treatment group will be compared with the results obtained for the concurrent vehicle control group from the same treatment regime using the Fisher's Exact Test. For the statistical analysis, results from replicate cultures will be combined to facilitate interpretation and maximize the power of statistical analysis.

6.3. Interpretation of Results

A Test Item is considered clearly **negative** if:

1. none of the Test Item concentrations selected for micronuclei scoring exhibit a statistically significant increase in the incidence of micronucleated binucleate cells compared to the concurrent Negative Control,
2. all results are inside the distribution of the historical negative control data (e.g. 95% control limits)

The Test Item is then considered unable to induce chromosome breaks and/or gain or loss in this test system.

A Test Item is considered to be clearly **positive** if:

1. at least one of the Test Item concentrations selected for detailed chromosome analysis exhibit a statistically significant increase in the incidence of micronucleated binucleate cells compared with the concurrent negative control ($p \leq 0.05$) at a concentration that does not greatly exceed a 50% cytoxicity level,
2. the increase is dose-related when evaluated with an appropriate trend test (a trend test will be performed when the incidence of micronucleated binucleate cells falls outside the distribution of the historical negative control database (e.g. 95% control limits)),
3. and the increase is outside the distribution of the historical negative control database (e.g. 95% control limits).

The Test Item is then considered able chromosome breaks and/or gain or loss in this test system.

An equivocal result is concluded if no definite judgment can be made to fit the above criteria. An equivocal result indicates that a definitive conclusion cannot be made by performing the in vitro micronucleus test under the conditions described in this Study Plan. Alternate testing conditions may be performed as an aid in evaluating the test results. Any additional testing or analysis will be approved by the sponsor and will be documented by Study Plan amendment.

7. RETENTION OF RECORDS

All study-specific raw data, documentation, Study Plan and Final Report from this study were archived at the Test Facility by no later than the date of Final Report issue, unless otherwise specified in the Study Plan. One year after issue of the unaudited Draft Report, the Sponsor will be contacted to determine the disposition of materials associated with the study.

Electronic data generated by the Test Facility were archived as noted above, except the reporting files stored on SDMS, which were archived at the Charles River Laboratories facility location in Wilmington, MA

8. RESULTS

8.1. Dose Formulation Analyses

Results of the dose formulation analysis are presented in Appendix 4.

All doses met acceptance criteria, with chemical analysis indicating achieved concentrations within ±10% of the theoretical concentration for the stock solution and ±15% for lower level solutions.

8.2. Dose Selection

The highest dose level tested was 500 µg/mL, the maximum dose level recommended by the ICH S2(R1) guideline.

In the absence of overt toxicity, the highest dose level of the Test Item selected for micronuclei scoring in each regime was the highest dose level tested (500 µg/mL). In addition, the next two lower dose levels were also subjected to examination (see Table 1).

8.3. Micronucleus Assessment

SM-102 did not cause any statistically significant increases in the incidence of micronucleated binucleate cells compared to the concurrent Negative Control at any experimental point (see Table 1 for summary and Appendix 3 for detailed results). In addition, the incidence of micronucleated binucleate cells for all Negative Control and Test Item groups was within the laboratory negative historical control range (see Appendix 5 for laboratory negative and positive historical control results).

The Positive Controls caused statistically significant increases in frequency of micronuclei in each regime of the study, confirming the sensitivity of the test system and the effectiveness of the S9 mix (see Appendix 5).

8.4. Incidental Observations

Precipitation was observed at the end of treatment at 500 µg/mL, in both the 4 hour regime and the 24 hour regime in the absence of S9 mix. No precipitation was observed at the end of treatment in the presence of S9 mix. Cloudy media was observed in the 4 hour regime in the absence of S9 mix at dose levels ≥ 93.3 µg/mL, in the 4 hour regime in the presence of S9 mix at the highest dose level and in the 24 hour regime at dose levels ≥ 286 µg/mL. No cytotoxicity was observed in the assay.

9. CONCLUSION

It is concluded that SM-102 did not show any evidence of genotoxic activity in this in vitro test for induction of micronuclei in human peripheral blood lymphocytes, when tested in accordance with regulatory guidelines.

Table 1 SM-102 - Summary Results and Statistical Analysis

Treatment	Conc. (µg/mL)	Average CBPI	(%) [a]Cytotoxicity	Total No. of BC examined	% MBC
4 hours treatment in the absence of S9 (0S9)					
Ethanol	-	1.8	0	2000	0.2
SM-102	163	1.9	-8	2000	0.5
	286	1.9	-8	2000	0.6
	500[ppt]	1.8	0	2000	0.1
NOC	0.25	1.5	43	1482	4.4**
4 hours treatment in the presence of S9 (+S9)					
Ethanol	-	1.8	0	2000	0.4
SM-102	163	1.8	-3	2000	0.5
	286	1.8	-1	2000	0.6
	500	1.8	1	2000	0.3
CP	10	1.4	46	2000	2.2**
24 hours treatment in the absence of S9 (0S9)					
Ethanol	-	1.7	0	2000	0.4
SM-102	163	1.7	-8	2000	0.2
	286	1.7	-2	2000	0.2
	500[ppt]	1.7	1	2000	0.3
MMC	0.10	1.6	3	2000	2.0**

CBPI = Cytokinesis-Block Proliferation Index.
a = Relative to the vehicle control.
BC = Binucleated cells.
MBC = Micronucleated binucleated cells.
NOC = Nocodazole
CP = Cyclophosphamide
MMC = Mitomycin C
* $p \leq 0.05$, ** $p \leq 0.01$ otherwise $p > 0.05$ Fisher's exact test with single-sided probabilities.
ppt = Precipitate visible in the culture medium at the end of treatment

Appendix 1
Study Plan, Amendment and Deviations

charles river

FINAL STUDY PLAN

Test Facility Study No. 9601568

SM-102
In Vitro Mammalian Cell Micronucleus Test in Human Peripheral Blood Lymphocytes

SPONSOR:
Moderna Therapeutics, Inc.
200 Technology Square, Third Floor
Cambridge, MA 02139, USA

TEST FACILITY:
Charles River Laboratories Montreal ULC
(b) (6)
[redacted]

08 September 2016

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1. OBJECTIVE

The objective of this study is to determine the potential genotoxicity of SM-102, using an in vitro mammalian cell micronucleus test in human peripheral blood lymphocytes.

2. PROPOSED STUDY SCHEDULE

Proposed study dates are listed below. Actual applicable dates will be included in the Final Report.

Experimental Start Date:	12 Sep 2016 (First date of study-specific data collection)
Experimental Completion Date:	13 Oct 2016 (Last date data are collected from the study)
Initiation of Dosing:	15 Sep 2016
Unaudited Draft Report:	27 Oct 2016
Final Report:	See Reporting Section 18

3. GUIDELINES FOR STUDY DESIGN

The design of this study was based on the study objective, the overall product development strategy for the Test Item, and the following study design guidelines:

- OECD Guideline 487. *OECD Guideline for the Testing of Chemicals – In Vitro Mammalian Cell Micronucleus Test.*
- ICH Harmonised Tripartite Guideline S2(R1). *Guidance on Genotoxicity Testing and Data Interpretation for Pharmaceuticals Intended for Human Use.*

4. REGULATORY COMPLIANCE

The study will be performed in accordance with the OECD Principles of Good Laboratory Practice and as accepted by Regulatory Authorities throughout the European Union, United States of America (FDA) and Japan (MHLW), and other countries that are signatories to the OECD Mutual Acceptance of Data Agreement.

Exceptions to GLPs include the following study elements:

- Characterization of the Test Item will be performed by the Sponsor or Sponsor subcontractor according to established SOPs, controls and approved test methodologies to ensure integrity and validity of the results generated; these analyses will not be conducted in compliance with the GLP or GMP regulations.

- Stability testing of the supplied Test Item will be performed by the Sponsor or Sponsor subcontractor at a laboratory that follows FDA Good Manufacturing Practice (GMP) regulations.

5. QUALITY ASSURANCE

5.1. Test Facility

The Test Facility Quality Assurance Program (QAP) will monitor the study to assure the facilities, equipment, personnel, methods, practices, records, and controls are in conformance with Good Laboratory Practice regulations. The QAP will review the Study Plan, conduct inspections at intervals adequate to assure the integrity of the study, and audit the Final Report to assure that it accurately describes the methods and standard operating procedures and that the reported results accurately reflect the raw data of the study.

6. SPONSOR

Sponsor Representative

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7. RESPONSIBLE PERSONNEL

Study Director

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E-mail: (b) (6)

Management Contact

(b) (6)
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Fax: (b) (6)
E-mail: (b) (6)

Individual Scientist (IS) at the Test Facility

Analytical Chemistry (b) (6)
Address as cited for Test Facility

Tel: (b) (6)
E-mail: (b) (6)

Each IS is required to report any deviations or other circumstances that could affect the quality or integrity of the study to the Study Director in a timely manner. Each IS will provide a report addressing their assigned phase of the study, which will be included as an appendix to the Final Report. The phase report will include the following:

- A listing of critical computerized systems used in the conduct and/or interpretation of the assigned study phase

8. TEST AND REFERENCE ITEMS

8.1. Test Item

Identification:	SM-102
Batch (Lot) No.:	RL-100-211-1
Retest Date:	To be included in the final report
Molecular Weight:	710.18
Purity:	95.3%; dose calculations will be corrected for purity
Storage Conditions:	Kept in a freezer set to maintain -20°C

8.2. Reference Items

8.2.1. Negative Control

Identification: Ethanol

Full details of the negative control, including supplier, lot number, storage, expiry date, will be maintained with the study raw data; appropriate details will be presented in the final report.

8.2.2. Positive Controls

In the Absence of S9 Mix:

Nocodazole (NOC), Mitomycin C (MMC)

In the Presence of S9 Mix:

Cyclophosphamide monohydrate (CP)

Full details of the positive controls, including supplier, lot number, storage, expiry date, formulation methods and concentrations (if applicable), will be maintained with the study raw data or as part of the Test Facility records; appropriate details will be presented in the final report.

8.3. Test Item Characterization

The Sponsor will provide to the Test Facility documentation of the identity, strength, purity, composition, and stability for the Test Item. A Certificate of Analysis or equivalent documentation will be provided for inclusion in the Final Report. The Sponsor will also provide information concerning the regulatory standard that was followed for these evaluations.

The Sponsor has appropriate documentation on file concerning the method of synthesis, fabrication or derivation of the Test Item, and this information is available to the appropriate regulatory agencies should it be requested.

8.4. Analysis of Test Item

The stability of the bulk Test Item will not be determined during the course of this study. Information to support the stability of each lot of bulk Test Item will be provided by the Sponsor.

8.5. Test Item Inventory and Disposition

Records of the receipt, distribution, storage and disposition of Test Item will be maintained. All unused Sponsor-supplied bulk Test Item will be returned to the Sponsor or discarded as instructed by the Sponsor following completion of the experimental phase of the study. Any remaining Reference Items (negative and positive controls) will be retained at the Test Facility or discarded upon expiry.

9. SAFETY

The safety precautions for the Test Item and dose formulations will be based on the information provided by the Sponsor either by an MSDS or similar document.

10. DOSE FORMULATION AND ANALYSIS

10.1. Preparation of Reference Items

10.1.1. Preparation of Negative Control

An adequate amount of the Negative Control, ethanol, will be dispensed into a vial for administration to control cultures. The aliquot will be stored in a refrigerator set to maintain 4°C until use. Any residual volumes will be discarded unless otherwise requested by the Study Director.

10.1.2. Positive Controls

Mitomycin C and Nocodazole may be prepared the day of use or up to 6 months prior to use; adequate amounts will be dispensed into vials, and stored in a freezer set to maintain -80°C (MMC) or -20°C (NOC), protected from light, until use. The aliquots will be removed from the freezer and allowed to warm to ambient room temperature before dosing. Cyclophosphamide will be prepared on the day of use. Any residual volumes will be discarded after completion of dosing.

10.2. Preparation of Test Item

Test Item dosing formulations will be prepared at appropriate concentrations to meet dosage level requirements. The Test Item will be prepared as a stock solution (50 mg/mL) in the vehicle (ethanol) and all lower level formulations will be prepared by serial dilution. The formulations will be prepared up to 4 days prior to use and will be stored in a refrigerator set to maintain 4°C until use. The formulations will be removed from the refrigerator and allowed to warm to room temperature for at least 30 minutes before dosing. Any residual volumes will be discarded unless otherwise requested by the Study Director.

10.3. Sample Collection and Analysis

Dose formulation samples will be collected from the dosing vial for analysis as indicated in the following table. Additional samples may be collected and analyzed at the discretion of the Study Director.

The Positive Controls formulations will not be subjected to analysis for safety reasons and because the biological response of the test system is considered to be the best measure of the appropriateness of the formulations.

Dose Formulation Sample Collection Schedule

Concentration
Dose numbers 1 to 10

All samples to be analyzed will be transferred under ambient conditions to the analytical laboratory at the Test Facility. Any residual/retained analytical samples will be discarded before issuance of the Final Report.

10.3.1. Analytical Method

Analyses described below will be performed by HPLC using a validated analytical procedure (Test Facility under Study Number 1801841).

10.3.1.1. Concentration Analysis

Samples for Analysis:	Duplicate samples, sent for analysis as noted in Section 10.3
Backup Samples:	Duplicate samples, maintained at the Test Facility. Backup samples may be analyzed at the discretion of the Study Director.
Sample Volume:	1 mL for dose numbers 1 and 2, 0.5 mL for dose numbers 3 to 10
Sampling Containers:	Appropriate sized glass containers
Storage Conditions:	Kept in a refrigerator set to maintain 4°C

Acceptance Criteria: Chemically analyzed mean values are normally expected to be within ±10% of theoretical concentration for stock solutions and ±15% for lower level solutions. The potential impact of any departure outside these limits will be addressed in the report.

10.3.1.2. Stability Analysis

Stability analyses performed previously at the Test Facility under Study Number 1801841 demonstrated that the Test Item is stable in the vehicle when prepared and stored under the same conditions at concentrations bracketing those used in the present study. Stability data have been retained in the study records for Study No. 1801841.

11. TEST SYSTEM

Primary cultures of human peripheral lymphocytes are preferred because of their low and stable background frequency of micronucleus formation compared with cell lines. In addition, human cells are generally the most relevant for risk assessment.

12. REASON FOR DOSE LEVEL SELECTION

Typically, the test item is dosed at a range of concentrations but is only assessed at the highest levels below the toxic level or, if non-toxic, at levels up to the standard limit of 0.5 mg/mL or 1 mM, whichever is lower. Compounds with limited aqueous solubility are tested up to a level expected to show visible precipitation in the culture medium at the end of treatment.

13. EXPERIMENTAL DESIGN

Experimental Design

Dose No.	Formulation Conc. (µg/mL)[a]	Final Conc. (µg/mL)	Number of Cultures		
			4 Hours (0S9)	4 Hours (+S9)	24 Hours (0S9)[c]
Negative Control	-	-	2	2	2
1/ SM-102	325	3.25	2	2	2
2/ SM-102	568	5.68	2	2	2
3/ SM-102	995	9.95	2	2	2
4/ SM-102	1740	17.4	2	2	2
5/ SM-102	3050	30.5	2	2	2
6/ SM-102	5330	53.3	2	2	2
7/ SM-102	9330	93.3	2	2	2
8/ SM-102	16300	163	2	2	2
9/ SM-102	28600	286	2	2	2
10/ SM-102	50000	500	2	2	2
NOC	25	0.25	2	-	-
	30	0.30	2	-	-
CP	1000	10	-	2	-
	1500	15	-	2	-
MMC	10	0.10	-	-	2
	20	0.20	-	-	2

a Theoretical concentrations; actual concentrations may differ slightly due to the limitations of the instruments used.

b Where the high level = 0.5 mg/mL

c The 24 hour incubation ('confirmatory phase') is required by regulatory authorities where the initial phase does not show any indication of genotoxicity for the Test Item; however, it is routinely performed at the same time as the initial phase to minimize potential delays in performance of the study.

14. EXPERIMENTAL PROCEDURES

14.1. Test System

14.1.1. Blood Sampling

A peripheral blood sample will be taken from young (approximately 18-35 years of age), healthy, non-smoking, male donor(s) with no known recent exposures to genotoxic chemicals or radiation, into tubes containing sodium heparin and will be held at room temperature for up to 2 hours prior to addition to the culture medium.

14.1.2. Culture Medium

Complete RPMI 1640 medium will be prepared by supplementing RPMI 1640 medium with the following filter-sterilized components: 10% (v/v) fetal bovine serum, 50 µg gentamycin per mL and 4 units heparin per mL.

14.1.3. Lymphocyte Culture

Phytohemagglutinin (PHA) will be added to whole blood mixed with medium to stimulate lymphocyte division. Aliquots of cell suspension will be placed in an incubator set to maintain 37°C with 5% CO_2 in a humidified atmosphere.

14.2. S9 Mix

The S9 mix, used as a model of mammalian metabolism, will be prepared on the day of use and will contain 10% v/v S9 fraction (Aroclor 1254 induced male rat liver fraction supplied by Moltox) and the following sterile cofactors: 8 mM $MgCl_2$, 33 mM KCl, 100 mM sodium phosphate buffer pH 7.4, 5 mM glucose-6-phosphate and 4 mM NADP. The S9 mix will be stored in a refrigerator set to maintain 4°C or on ice until required. A copy of the manufacturer's quality control certificate(s) for the S9 fraction will be retained as raw data.

14.3. Treatment

Treatments will be performed approximately 48 hours after culture initiation. Appropriate dilutions of Test Item and Positive Control formulations will be prepared so as to reach the final concentrations indicated in the experimental design. Where the dosing volume is greater than 50 µL per mL, medium will be removed from the culture to ensure achievement of the nominal final concentration.

The test item will be tested over a wide range of dose levels so that analyzable cells will be available for at least three dose levels for each treatment regime (see experimental design) in the event of toxicity.

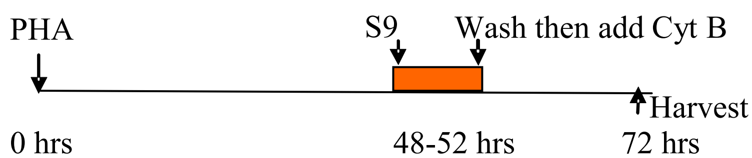
Cultures tested in the absence of S9 mix will be treated as indicated in the study design then returned to the incubator for 4 or 24 hours as appropriate. For cultures tested in the presence of S9 mix, 0.2 mL of S9 mix per mL of culture will be added immediately prior to treatment as indicated in the experimental design. Cultures will be returned to the incubator for 4 hours.

14.4. Medium Change (Wash) and Cytochalasin B (Cyt B) Treatment

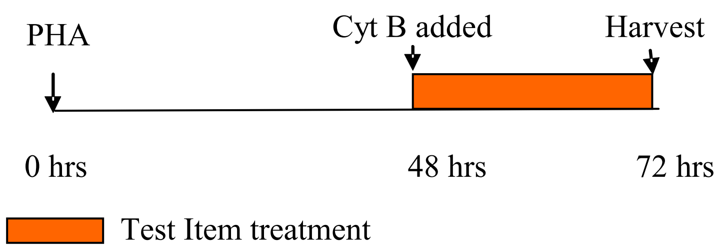
After the 4-hour treatments, cultures will be centrifuged, and the supernatant replaced with fresh complete medium containing 6 µg Cytochalasin B per mL. Incubation will be continued for a further 20 hours prior to harvesting.

For the 24-hour treatment, the medium added to the cultures just prior to dosing will contain 6 µg Cytochalasin B per mL of culture medium. The 24-hour treatment cultures will be harvested at the end of the treatment period.

4-hour treatment in the absence or presence of S9 mix (+S9)



24-hour treatment in the absence of S9 (0S9)



14.5. Harvesting and Staining

Cultures will be harvested by centrifugation. The cell pellet will be resuspended in hypotonic solution (0.075 M KCl) at ambient temperature. Fixative (24:1 v/v, methanol: acetic acid) will be mixed with the suspended cells and following another centrifugation, the cells will be treated with 3 changes of fixative. After the third change of fixative, the cell pellet will be resuspended in fixative at an appropriate density for slide preparation. The fixed cells will be dropped onto clean slides and air-dried before staining. At least one slide will be prepared from each culture. Fixed cells not used for slide preparation will be discarded. Selected slides will be stained with the fluorescent metachromatic dye, acridine orange. Stained fixed slides will be kept for potential retrospective examination until study finalization.

15. SLIDE EXAMINATION

15.1. Cytokinesis-Block Proliferation Index (CBPI)

Slides will be examined visually for toxicity and slides from a minimum of 3 concentrations up to the toxic concentration, or the lowest concentration which results in a visible precipitate in the culture medium at the end of treatment, whichever is least (or top concentration if there is no cytotoxicity or precipitate limitations) will be selected for CBPI determination. Selected slides will be randomized using Microsoft® Excel then encoded to minimize potential operator bias.

The CBPI will be determined by examination of at least 500 cells (if available) per culture. Lymphocyte toxicity is normally indicated by a decreased CBPI compared to the concurrent control group.

$$\text{CBPI} = \frac{((\text{No. mononucleate cells}) + (2 \times \text{No. binucleate cells}) + (3 \times \text{No. multinucleate cells}))}{\text{Total number of cells}}$$

% Cytotoxicity = $100\text{-}100((\text{CBPI}_T\text{-}1) \div (\text{CBPI}_C\text{-}1))$ where

T= test item treatment group

C= Negative Control group

15.2. Selection of Slides for Micronucleus Assessment

Justification for selection of concentration levels for examination will be presented in the results section of the report. Routinely, slides for examination for micronuclei are selected primarily on the basis of the CBPI results. Normally, the highest level examined is the lowest concentration which results in approximately 55% toxicity based on CBPI with a sufficient number of scorable binucleate cells or the lowest concentration which causes precipitation at the end of treatment, whichever is least; in the absence of toxicity or precipitation, it will be the highest dose tested. In addition, at least two lower dose levels, the Negative Control and one dose level of the positive controls for each regime will be included in the detailed analysis.

15.3. Microscopic Analysis of Micronucleus Frequencies

Slides selected for analysis of micronucleus frequencies will be examined by fluorescence microscopy using a blue excitation filter and a yellow barrier filter, and (where practical) a total of 2000 binucleate cells per experimental point (1000 per culture) will be examined for the presence of micronuclei using oil-immersion optics.

Readable binucleate cells are identified by the following criteria:

- The cell must have two main nuclei
- The two main nuclei must each have an intact and well-defined membrane
- The two main nuclei must be contained within the cytoplasm
- The cell must be visible in its entirety in the field
- The area around the cell must not contain micronucleus-like debris
- The cytoplasmic boundary should be intact and distinguishable from the boundaries of adjacent cells

Micronuclei (MN) are identified by the following criteria:

- The diameter of the MN must not exceed $1/3^{rd}$ of each of the two main nuclei diameter

- The micronuclei can touch but must not overlap the two main nuclei
- Micronuclei should be large enough to discern morphological characteristics
- Micronuclei should possess a generally rounded shape with a clearly defined outline
- Micronuclei should be similar in color to the nuclei
- Should lie in the same focal plane as the cell
- Micronuclei must not be linked to the nuclei by a nucleoplasmic bridge
- Micronuclei must be within cytoplasmic boundary
- Micronuclei must be non-refractive (staining)

The location (Vernier readings) of the first nine micronucleated binucleate cells (MBC) will be recorded for potential peer review. The % micronucleated binucleate cells (% MBC) is the proportion of micronucleated binucleate cells over the total number of binucleate cells evaluated. All slides will be discarded after completion of the experimental phase of study.

16. COMPUTERIZED SYSTEMS

The following proposed critical computerized systems may be used in the study. The actual critical computerized systems used will be specified in the Final Report.

Data for parameters not required by Study Plan, which are automatically generated by analytical devices used will be retained on file but not reported. Statistical analysis results that are generated by the program but are not required by Study Plan and/or are not scientifically relevant will be retained on file but will not be included in the tabulations.

Proposed Critical Computerized Systems

System Name	Description of Data Collected and/or Analyzed
Excel (Microsoft)	Slide randomization and arithmetic calculations of mean values, etc, for presentation in reports.
SAS	Statistical analyses of number of micronucleus cells
Mesa Laboratories AmegaView CMS	Continuous Monitoring System. Monitoring of standalone fridges, freezers, incubators, and selected laboratories to measure temperature, relative humidity, and CO_2, as appropriate.
Johnson Controls Metasys	Building Automation System. Control of HVAC and other building systems, as well as temperature/humidity control and trending in selected laboratories.
Empower 3 (Waters Corporation)	Dose formulation analyses using HPLC.
Empower 3 and Microsoft Excel	Regression analysis and descriptive statistics for dose formulation analytical data.

17. EVALUATION AND INTERPRETATION OF RESULTS

17.1. Assay acceptance criteria:

- *Acceptable Negative Control:* The incidence of micronucleated binucleate cells in the Negative Control should be considered acceptable for addition to the negative control data base (should be within the 95% control limits of the distribution of the negative control database or, where the incidence of micronucleated binucleate cells falls outside of the limits, should not be an extreme outlier and there is evidence that the test system is under control).
- *Acceptable Positive Control:* The number of micronucleated binucleate cells in the Positive Control cultures should be compatible with those generated in the historical positive control database and produce a statistically significant increase compared to the concurrent Negative Control.
- *Acceptable Cell Proliferation*: Cell proliferation, as measured by the CPBI, should indicate that the treatments are conducted at appropriate levels of cytotoxicity.
- *Experimental Conditions:* All three treatment regimes are to be used in the assay unless a positive result is obtained in one of the regimes.
- *Acceptable Number of Cells and Analyzable Concentrations:* An acceptable number of cells (see Section 15) and at least three test concentrations are obtained.
- *Selection of Top Concentration:* The criteria for the selection of the top concentration are consistent with Section 15.2.

In the event that the controls fall slightly outside the normal range (historical or Study Plan), the Study Director will be allowed discretion to accept the results of the experiment as valid based on the biological significance.

17.2. Statistical Analysis

The results obtained for each treatment group will be compared with the results obtained for the concurrent Negative Control group from the same treatment regime using the Fisher's Exact Test. For the statistical analysis, results from replicate cultures will be combined to facilitate interpretation and maximize the power of statistical analysis.

17.3. Interpretation of Results

A Test Item is considered clearly **negative** if:

1. none of the Test Item concentrations selected for micronuclei scoring exhibit a statistically significant increase in the incidence of micronucleated binucleate cells compared to the concurrent Negative Control,

2. all results are inside the distribution of the historical negative control data (e.g. 95% control limits)

The Test Item is then considered unable to induce chromosome breaks and/or gain or loss in this test system.

A Test Item is considered to be clearly **positive** if:

1. at least one of the Test Item concentrations selected for detailed chromosome analysis exhibit a statistically significant increase in the incidence of micronucleated binucleate cells compared with the concurrent Negative Control ($p \leq 0.05$) at a concentration that does not greatly exceed a 55% cytoxicity,
2. the increase is dose-related when evaluated with an appropriate trend test (a trend test will be performed when the incidence of micronucleated binucleate cells falls outside the distribution of the historical negative control database (e.g. 95% control limits)),
3. and the increase is outside the distribution of the historical negative control database (e.g. 95% control limits).

The Test Item is then considered able chromosome breaks and/or gain or loss in this test system.

An equivocal result is concluded if no definite judgment can be made to fit the above criteria. An equivocal result indicates that a definitive conclusion cannot be made by performing the in vitro micronucleus test under the conditions described in this Study Plan. Alternate testing conditions may be performed as an aid in evaluating the test results. Any additional testing or analysis will be approved by the sponsor and will be documented by Study Plan amendment

18. AMENDMENTS AND DEVIATIONS

Changes to the approved Study Plan shall be made in the form of an amendment, which will be signed and dated by the Study Director. Every reasonable effort will be made to discuss any necessary Study Plan changes in advance with the Sponsor.

All Study Plan and SOP deviations will be documented in the study records. The Study Director will notify the Sponsor of deviations that may result in a significant impact on the study as soon as possible.

19. RETENTION OF RECORDS

All study-specific raw data, electronic data, documentation, study plan, retained samples, and interim (if applicable) and final reports will be archived by no later than the date of final report issue. All materials generated by Charles River from this study will be transferred to a Charles River archive. One year after issue of the unaudited draft report, the Sponsor will be contacted to determine the disposition of materials associated with the study.

Records to be maintained will include, but will not be limited to, documentation and data for the following:

- Study Plan, Study Plan amendments, and deviations
- Study-related correspondence
- Test system receipt (or collection) and storage
- Test and Reference Items receipt, identification, preparation, and analysis
- On-study measurements and observations
- Statistical analysis results

20. REPORTING

A comprehensive unaudited Draft Report will be prepared following completion of the study and will be finalized following consultation with the Sponsor. The report will include all information necessary to provide a complete and accurate description of the experimental methods and results and any circumstances that may have affected the quality or integrity of the study.

The Sponsor will receive an electronic version of the unaudited Draft Report provided in Microsoft Word format and an electronic version of the Final Report provided in Adobe Acrobat PDF format (hyperlinked and searchable). The PDF document will be created from native electronic files to the every possible extend, including text and tables generated by the Test Facility. Report components not available in native electronic files and/or original signature pages will be scanned and converted to PDF image files for incorporation.

Reports should be finalized within 6 months of issuance of the unaudited Draft Report. If the Sponsor has not provided comments to the report within 6 months of draft issuance, the report will be finalized by the Test Facility unless other arrangements are made by the Sponsor.

TEST FACILITY APPROVAL

The signature below acknowledges Test Facility Management's responsibility to the study as defined by the relevant GLP regulations.

(b) (6) ______________________ Date: 08 Sep 2016

The signature below indicates that the Study Director approves the study plan.

(b) (6) ______________________ Date: 08 Sep 2016

SPONSOR APPROVAL

The study plan was approved by the Sponsor by email on 07 Sep 2016. The signature below confirms the approval of the study plan by the Sponsor Representative.

______________(b) (6)_______________________Date: 15-Sep-16

(b) (6)

STUDY PLAN AMENDMENT NO. 1

Test Facility Study No. 9601568

SM-102
In Vitro Mammalian Cell Micronucleus Test in Human Peripheral Blood Lymphocytes

SPONSOR:
Moderna Therapeutics, Inc.
200 Technology Square, Third Floor
Cambridge, MA 02139, USA

TEST FACILITY:
Charles River Laboratories Montreal ULC
(b) (6) [redacted]

20 December 2016

Page 1 of 18

SUMMARY OF CHANGES AND JUSTIFICATIONS

Study Plan effective date: Date: 08 Sep 2016

Note: When applicable, additions are indicated in bold underlined text and deletions are indicated in bold strikethrough text in the affected sections of the document.

Section	**Justification**
Amendment 1	**Date: 20 Dec 2016**
16. Computerized Systems	Provantis added as a critical computerized system for statistical analysis

TABLE OF CONTENTS

1. OBJECTIVE

The objective of this study is to determine the potential genotoxicity of SM-102, using an in vitro mammalian cell micronucleus test in human peripheral blood lymphocytes.

2. PROPOSED STUDY SCHEDULE

Proposed study dates are listed below. Actual applicable dates will be included in the Final Report.

Experimental Start Date:	12 Sep 2016 (First date of study-specific data collection)
Experimental Completion Date:	13 Oct 2016 (Last date data are collected from the study)
Initiation of Dosing:	15 Sep 2016
Unaudited Draft Report:	27 Oct 2016
Final Report:	See Reporting Section 18

3. GUIDELINES FOR STUDY DESIGN

The design of this study was based on the study objective, the overall product development strategy for the Test Item, and the following study design guidelines:

- OECD Guideline 487. *OECD Guideline for the Testing of Chemicals – In Vitro Mammalian Cell Micronucleus Test.*
- ICH Harmonised Tripartite Guideline S2(R1). *Guidance on Genotoxicity Testing and Data Interpretation for Pharmaceuticals Intended for Human Use.*

4. REGULATORY COMPLIANCE

The study will be performed in accordance with the OECD Principles of Good Laboratory Practice and as accepted by Regulatory Authorities throughout the European Union, United States of America (FDA) and Japan (MHLW), and other countries that are signatories to the OECD Mutual Acceptance of Data Agreement.

Exceptions to GLPs include the following study elements:

- Characterization of the Test Item will be performed by the Sponsor or Sponsor subcontractor according to established SOPs, controls and approved test methodologies to ensure integrity and validity of the results generated; these analyses will not be conducted in compliance with the GLP or GMP regulations.

- Stability testing of the supplied Test Item will be performed by the Sponsor or Sponsor subcontractor at a laboratory that follows FDA Good Manufacturing Practice (GMP) regulations.

5. QUALITY ASSURANCE

5.1. Test Facility

The Test Facility Quality Assurance Program (QAP) will monitor the study to assure the facilities, equipment, personnel, methods, practices, records, and controls are in conformance with Good Laboratory Practice regulations. The QAP will review the Study Plan, conduct inspections at intervals adequate to assure the integrity of the study, and audit the Final Report to assure that it accurately describes the methods and standard operating procedures and that the reported results accurately reflect the raw data of the study.

6. SPONSOR

Sponsor Representative

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7. RESPONSIBLE PERSONNEL

Study Director

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Fax: (b) (6)
E-mail: (b) (6)

Individual Scientist (IS) at the Test Facility

Analytical Chemistry (b) (6)
Address as cited for Test Facility

Tel: (b) (6)
E-mail: (b) (6)

Each IS is required to report any deviations or other circumstances that could affect the quality or integrity of the study to the Study Director in a timely manner. Each IS will provide a report addressing their assigned phase of the study, which will be included as an appendix to the Final Report. The phase report will include the following:

- A listing of critical computerized systems used in the conduct and/or interpretation of the assigned study phase

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8.1. Test Item

Identification: SM-102

Batch (Lot) No.:	RL-100-211-1
Retest Date:	To be included in the final report
Molecular Weight:	710.18
Purity:	95.3%; dose calculations will be corrected for purity
Storage Conditions:	Kept in a freezer set to maintain -20°C

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8.2.1. Negative Control

Identification: Ethanol

Full details of the negative control, including supplier, lot number, storage, expiry date, will be maintained with the study raw data; appropriate details will be presented in the final report.

8.2.2. Positive Controls

In the Absence of S9 Mix:

Nocodazole (NOC), Mitomycin C (MMC)

In the Presence of S9 Mix:

Cyclophosphamide monohydrate (CP)

Full details of the positive controls, including supplier, lot number, storage, expiry date, formulation methods and concentrations (if applicable), will be maintained with the study raw data or as part of the Test Facility records; appropriate details will be presented in the final report.

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The stability of the bulk Test Item will not be determined during the course of this study. Information to support the stability of each lot of bulk Test Item will be provided by the Sponsor.

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Records of the receipt, distribution, storage and disposition of Test Item will be maintained. All unused Sponsor-supplied bulk Test Item will be returned to the Sponsor or discarded as instructed by the Sponsor following completion of the experimental phase of the study. Any remaining Reference Items (negative and positive controls) will be retained at the Test Facility or discarded upon expiry.

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The safety precautions for the Test Item and dose formulations will be based on the information provided by the Sponsor either by an MSDS or similar document.

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10.1. Preparation of Reference Items

10.1.1. Preparation of Negative Control

An adequate amount of the Negative Control, ethanol, will be dispensed into a vial for administration to control cultures. The aliquot will be stored in a refrigerator set to maintain 4°C until use. Any residual volumes will be discarded unless otherwise requested by the Study Director.

10.1.2. Positive Controls

Mitomycin C and Nocodazole may be prepared the day of use or up to 6 months prior to use; adequate amounts will be dispensed into vials, and stored in a freezer set to maintain -80°C (MMC) or -20°C (NOC), protected from light, until use. The aliquots will be removed from the freezer and allowed to warm to ambient room temperature before dosing. Cyclophosphamide will be prepared on the day of use. Any residual volumes will be discarded after completion of dosing.

10.2. Preparation of Test Item

Test Item dosing formulations will be prepared at appropriate concentrations to meet dosage level requirements. The Test Item will be prepared as a stock solution (50 mg/mL) in the vehicle (ethanol) and all lower level formulations will be prepared by serial dilution. The formulations will be prepared up to 4 days prior to use and will be stored in a refrigerator set to maintain 4°C until use. The formulations will be removed from the refrigerator and allowed to warm to room temperature for at least 30 minutes before dosing. Any residual volumes will be discarded unless otherwise requested by the Study Director.

10.3. Sample Collection and Analysis

Dose formulation samples will be collected from the dosing vial for analysis as indicated in the following table. Additional samples may be collected and analyzed at the discretion of the Study Director.

The Positive Controls formulations will not be subjected to analysis for safety reasons and because the biological response of the test system is considered to be the best measure of the appropriateness of the formulations.

Dose Formulation Sample Collection Schedule

Concentration
Dose numbers 1 to 10

All samples to be analyzed will be transferred under ambient conditions to the analytical laboratory at the Test Facility. Any residual/retained analytical samples will be discarded before issuance of the Final Report.

10.3.1. Analytical Method

Analyses described below will be performed by HPLC using a validated analytical procedure (Test Facility under Study Number 1801841).

10.3.1.1. Concentration Analysis

Samples for Analysis:	Duplicate samples, sent for analysis as noted in Section 10.3
Backup Samples:	Duplicate samples, maintained at the Test Facility. Backup samples may be analyzed at the discretion of the Study Director.
Sample Volume:	1 mL for dose numbers 1 and 2, 0.5 mL for dose numbers 3 to 10
Sampling Containers:	Appropriate sized glass containers
Storage Conditions:	Kept in a refrigerator set to maintain 4°C

Acceptance Criteria: Chemically analyzed mean values are normally expected to be within ±10% of theoretical concentration for stock solutions and ±15% for lower level solutions. The potential impact of any departure outside these limits will be addressed in the report.

10.3.1.2. Stability Analysis

Stability analyses performed previously at the Test Facility under Study Number 1801841 demonstrated that the Test Item is stable in the vehicle when prepared and stored under the same conditions at concentrations bracketing those used in the present study. Stability data have been retained in the study records for Study No. 1801841.

11. TEST SYSTEM

Primary cultures of human peripheral lymphocytes are preferred because of their low and stable background frequency of micronucleus formation compared with cell lines. In addition, human cells are generally the most relevant for risk assessment.

12. REASON FOR DOSE LEVEL SELECTION

Typically, the test item is dosed at a range of concentrations but is only assessed at the highest levels below the toxic level or, if non-toxic, at levels up to the standard limit of 0.5 mg/mL or 1 mM, whichever is lower. Compounds with limited aqueous solubility are tested up to a level expected to show visible precipitation in the culture medium at the end of treatment.

13. EXPERIMENTAL DESIGN

Experimental Design

Dose No.	Formulation Conc. (µg/mL)[a]	Final Conc. (µg/mL)	Number of Cultures		
			4 Hours (0S9)	4 Hours (+S9)	24 Hours (0S9)[c]
Negative Control	-	-	2	2	2
1/ SM-102	325	3.25	2	2	2
2/ SM-102	568	5.68	2	2	2
3/ SM-102	995	9.95	2	2	2
4/ SM-102	1740	17.4	2	2	2
5/ SM-102	3050	30.5	2	2	2
6/ SM-102	5330	53.3	2	2	2
7/ SM-102	9330	93.3	2	2	2
8/ SM-102	16300	163	2	2	2
9/ SM-102	28600	286	2	2	2
10/ SM-102	50000	500	2	2	2
NOC	25	0.25	2	-	-
	30	0.30	2	-	-
CP	1000	10	-	2	-
	1500	15	-	2	-
MMC	10	0.10	-	-	2
	20	0.20	-	-	2

a Theoretical concentrations; actual concentrations may differ slightly due to the limitations of the instruments used.

b Where the high level = 0.5 mg/mL

c The 24 hour incubation ('confirmatory phase') is required by regulatory authorities where the initial phase does not show any indication of genotoxicity for the Test Item; however, it is routinely performed at the same time as the initial phase to minimize potential delays in performance of the study.

14. EXPERIMENTAL PROCEDURES

14.1. Test System

14.1.1. Blood Sampling

A peripheral blood sample will be taken from young (approximately 18-35 years of age), healthy, non-smoking, male donor(s) with no known recent exposures to genotoxic chemicals or radiation, into tubes containing sodium heparin and will be held at room temperature for up to 2 hours prior to addition to the culture medium.

14.1.2. Culture Medium

Complete RPMI 1640 medium will be prepared by supplementing RPMI 1640 medium with the following filter-sterilized components: 10% (v/v) fetal bovine serum, 50 µg gentamycin per mL and 4 units heparin per mL.

14.1.3. Lymphocyte Culture

Phytohemagglutinin (PHA) will be added to whole blood mixed with medium to stimulate lymphocyte division. Aliquots of cell suspension will be placed in an incubator set to maintain 37°C with 5% CO_2 in a humidified atmosphere.

14.2. S9 Mix

The S9 mix, used as a model of mammalian metabolism, will be prepared on the day of use and will contain 10% v/v S9 fraction (Aroclor 1254 induced male rat liver fraction supplied by Moltox) and the following sterile cofactors: 8 mM $MgCl_2$, 33 mM KCl, 100 mM sodium phosphate buffer pH 7.4, 5 mM glucose-6-phosphate and 4 mM NADP. The S9 mix will be stored in a refrigerator set to maintain 4°C or on ice until required. A copy of the manufacturer's quality control certificate(s) for the S9 fraction will be retained as raw data.

14.3. Treatment

Treatments will be performed approximately 48 hours after culture initiation. Appropriate dilutions of Test Item and Positive Control formulations will be prepared so as to reach the final concentrations indicated in the experimental design. Where the dosing volume is greater than 50 µL per mL, medium will be removed from the culture to ensure achievement of the nominal final concentration.

The test item will be tested over a wide range of dose levels so that analyzable cells will be available for at least three dose levels for each treatment regime (see experimental design) in the event of toxicity.

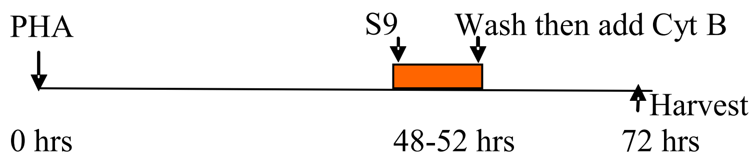
Cultures tested in the absence of S9 mix will be treated as indicated in the study design then returned to the incubator for 4 or 24 hours as appropriate. For cultures tested in the presence of S9 mix, 0.2 mL of S9 mix per mL of culture will be added immediately prior to treatment as indicated in the experimental design. Cultures will be returned to the incubator for 4 hours.

14.4. Medium Change (Wash) and Cytochalasin B (Cyt B) Treatment

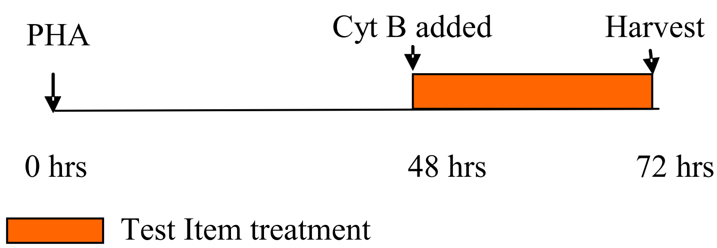
After the 4-hour treatments, cultures will be centrifuged, and the supernatant replaced with fresh complete medium containing 6 µg Cytochalasin B per mL. Incubation will be continued for a further 20 hours prior to harvesting.

For the 24-hour treatment, the medium added to the cultures just prior to dosing will contain 6 µg Cytochalasin B per mL of culture medium. The 24-hour treatment cultures will be harvested at the end of the treatment period.

4-hour treatment in the absence or presence of S9 mix (+S9)



24-hour treatment in the absence of S9 (0S9)



14.5. Harvesting and Staining

Cultures will be harvested by centrifugation. The cell pellet will be resuspended in hypotonic solution (0.075 M KCl) at ambient temperature. Fixative (24:1 v/v, methanol: acetic acid) will be mixed with the suspended cells and following another centrifugation, the cells will be treated with 3 changes of fixative. After the third change of fixative, the cell pellet will be resuspended in fixative at an appropriate density for slide preparation. The fixed cells will be dropped onto clean slides and air-dried before staining. At least one slide will be prepared from each culture. Fixed cells not used for slide preparation will be discarded. Selected slides will be stained with the fluorescent metachromatic dye, acridine orange. Stained fixed slides will be kept for potential retrospective examination until study finalization.

15. SLIDE EXAMINATION

15.1. Cytokinesis-Block Proliferation Index (CBPI)

Slides will be examined visually for toxicity and slides from a minimum of 3 concentrations up to the toxic concentration, or the lowest concentration which results in a visible precipitate in the culture medium at the end of treatment, whichever is least (or top concentration if there is no cytotoxicity or precipitate limitations) will be selected for CBPI determination. Selected slides will be randomized using Microsoft® Excel then encoded to minimize potential operator bias.

The CBPI will be determined by examination of at least 500 cells (if available) per culture. Lymphocyte toxicity is normally indicated by a decreased CBPI compared to the concurrent control group.

$$\text{CBPI} = \frac{(\text{No. mononucleate cells}) + (2 \times \text{No. binucleate cells}) + (3 \times \text{No. multinucleate cells})}{\text{Total number of cells}}$$

% Cytotoxicity = 100-100(($CBPI_T$-1) ÷ ($CBPI_C$-1)) where

T= test item treatment group

C= Negative Control group

15.2. Selection of Slides for Micronucleus Assessment

Justification for selection of concentration levels for examination will be presented in the results section of the report. Routinely, slides for examination for micronuclei are selected primarily on the basis of the CBPI results. Normally, the highest level examined is the lowest concentration which results in approximately 55% toxicity based on CBPI with a sufficient number of scorable binucleate cells or the lowest concentration which causes precipitation at the end of treatment, whichever is least; in the absence of toxicity or precipitation, it will be the highest dose tested. In addition, at least two lower dose levels, the Negative Control and one dose level of the positive controls for each regime will be included in the detailed analysis.

15.3. Microscopic Analysis of Micronucleus Frequencies

Slides selected for analysis of micronucleus frequencies will be examined by fluorescence microscopy using a blue excitation filter and a yellow barrier filter, and (where practical) a total of 2000 binucleate cells per experimental point (1000 per culture) will be examined for the presence of micronuclei using oil-immersion optics.

Readable binucleate cells are identified by the following criteria:

- The cell must have two main nuclei
- The two main nuclei must each have an intact and well-defined membrane
- The two main nuclei must be contained within the cytoplasm
- The cell must be visible in its entirety in the field
- The area around the cell must not contain micronucleus-like debris
- The cytoplasmic boundary should be intact and distinguishable from the boundaries of adjacent cells

Micronuclei (MN) are identified by the following criteria:

- The diameter of the MN must not exceed 1/3rd of each of the two main nuclei diameter

- The micronuclei can touch but must not overlap the two main nuclei
- Micronuclei should be large enough to discern morphological characteristics
- Micronuclei should possess a generally rounded shape with a clearly defined outline
- Micronuclei should be similar in color to the nuclei
- Should lie in the same focal plane as the cell
- Micronuclei must not be linked to the nuclei by a nucleoplasmic bridge
- Micronuclei must be within cytoplasmic boundary
- Micronuclei must be non-refractive (staining)

The location (Vernier readings) of the first nine micronucleated binucleate cells (MBC) will be recorded for potential peer review. The % micronucleated binucleate cells (% MBC) is the proportion of micronucleated binucleate cells over the total number of binucleate cells evaluated. All slides will be discarded after completion of the experimental phase of study.

16. COMPUTERIZED SYSTEMS

The following proposed critical computerized systems may be used in the study. The actual critical computerized systems used will be specified in the Final Report.

Data for parameters not required by Study Plan, which are automatically generated by analytical devices used will be retained on file but not reported. Statistical analysis results that are generated by the program but are not required by Study Plan and/or are not scientifically relevant will be retained on file but will not be included in the tabulations.

Proposed Critical Computerized Systems

System Name	Description of Data Collected and/or Analyzed
Excel (Microsoft)	Slide randomization and arithmetic calculations of mean values, etc, for presentation in reports.
SAS	Statistical analyses of number of micronucleus cells
Provantis	**Statistical analyses of the number of micronucleated cells where less than 2000 binucleated cells were examined per experimental point**
Mesa Laboratories AmegaView CMS	Continuous Monitoring System. Monitoring of standalone fridges, freezers, incubators, and selected laboratories to measure temperature, relative humidity, and CO_2, as appropriate.
Johnson Controls Metasys	Building Automation System. Control of HVAC and other building systems, as well as temperature/humidity control and trending in selected laboratories.
Empower 3 (Waters Corporation)	Dose formulation analyses using HPLC.
Empower 3 and Microsoft Excel	Regression analysis and descriptive statistics for dose formulation analytical data.

17. EVALUATION AND INTERPRETATION OF RESULTS

17.1. Assay acceptance criteria:

- *Acceptable Negative Control:* The incidence of micronucleated binucleate cells in the Negative Control should be considered acceptable for addition to the negative control data base (should be within the 95% control limits of the distribution of the negative control database or, where the incidence of micronucleated binucleate cells falls outside of the limits, should not be an extreme outlier and there is evidence that the test system is under control).
- *Acceptable Positive Control:* The number of micronucleated binucleate cells in the Positive Control cultures should be compatible with those generated in the historical positive control database and produce a statistically significant increase compared to the concurrent Negative Control.
- *Acceptable Cell Proliferation*: Cell proliferation, as measured by the CPBI, should indicate that the treatments are conducted at appropriate levels of cytotoxicity.
- *Experimental Conditions:* All three treatment regimes are to be used in the assay unless a positive result is obtained in one of the regimes.
- *Acceptable Number of Cells and Analyzable Concentrations:* An acceptable number of cells (see Section 15) and at least three test concentrations are obtained.
- *Selection of Top Concentration:* The criteria for the selection of the top concentration are consistent with Section 15.2.

In the event that the controls fall slightly outside the normal range (historical or Study Plan), the Study Director will be allowed discretion to accept the results of the experiment as valid based on the biological significance.

17.2. Statistical Analysis

The results obtained for each treatment group will be compared with the results obtained for the concurrent Negative Control group from the same treatment regime using the Fisher's Exact Test. For the statistical analysis, results from replicate cultures will be combined to facilitate interpretation and maximize the power of statistical analysis.

17.3. Interpretation of Results

A Test Item is considered clearly **negative** if:

1. none of the Test Item concentrations selected for micronuclei scoring exhibit a statistically significant increase in the incidence of micronucleated binucleate cells compared to the concurrent Negative Control,

2. all results are inside the distribution of the historical negative control data (e.g. 95% control limits)

The Test Item is then considered unable to induce chromosome breaks and/or gain or loss in this test system.

A Test Item is considered to be clearly **positive** if:

1. at least one of the Test Item concentrations selected for detailed chromosome analysis exhibit a statistically significant increase in the incidence of micronucleated binucleate cells compared with the concurrent Negative Control ($p \leq 0.05$) at a concentration that does not greatly exceed a 55% cytoxicity,
2. the increase is dose-related when evaluated with an appropriate trend test (a trend test will be performed when the incidence of micronucleated binucleate cells falls outside the distribution of the historical negative control database (e.g. 95% control limits)),
3. and the increase is outside the distribution of the historical negative control database (e.g. 95% control limits).

The Test Item is then considered able chromosome breaks and/or gain or loss in this test system.

An equivocal result is concluded if no definite judgment can be made to fit the above criteria. An equivocal result indicates that a definitive conclusion cannot be made by performing the in vitro micronucleus test under the conditions described in this Study Plan. Alternate testing conditions may be performed as an aid in evaluating the test results. Any additional testing or analysis will be approved by the sponsor and will be documented by Study Plan amendment

18. AMENDMENTS AND DEVIATIONS

Changes to the approved Study Plan shall be made in the form of an amendment, which will be signed and dated by the Study Director. Every reasonable effort will be made to discuss any necessary Study Plan changes in advance with the Sponsor.

All Study Plan and SOP deviations will be documented in the study records. The Study Director will notify the Sponsor of deviations that may result in a significant impact on the study as soon as possible.

19. RETENTION OF RECORDS

All study-specific raw data, electronic data, documentation, study plan, retained samples, and interim (if applicable) and final reports will be archived by no later than the date of final report issue. All materials generated by Charles River from this study will be transferred to a Charles River archive. One year after issue of the unaudited draft report, the Sponsor will be contacted to determine the disposition of materials associated with the study.

Records to be maintained will include, but will not be limited to, documentation and data for the following:

- Study Plan, Study Plan amendments, and deviations
- Study-related correspondence
- Test system receipt (or collection) and storage
- Test and Reference Items receipt, identification, preparation, and analysis
- On-study measurements and observations
- Statistical analysis results

20. REPORTING

A comprehensive unaudited Draft Report will be prepared following completion of the study and will be finalized following consultation with the Sponsor. The report will include all information necessary to provide a complete and accurate description of the experimental methods and results and any circumstances that may have affected the quality or integrity of the study.

The Sponsor will receive an electronic version of the unaudited Draft Report provided in Microsoft Word format and an electronic version of the Final Report provided in Adobe Acrobat PDF format (hyperlinked and searchable). The PDF document will be created from native electronic files to the every possible extend, including text and tables generated by the Test Facility. Report components not available in native electronic files and/or original signature pages will be scanned and converted to PDF image files for incorporation.

Reports should be finalized within 6 months of issuance of the unaudited Draft Report. If the Sponsor has not provided comments to the report within 6 months of draft issuance, the report will be finalized by the Test Facility unless other arrangements are made by the Sponsor.

AMENDMENT APPROVAL

(b) (6) ____________________ Date: 20 Dec 2016

DEVIATIONS

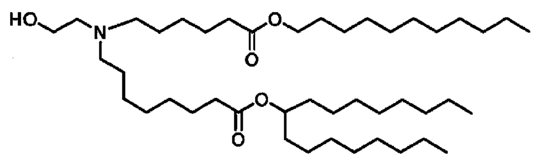
All deviations that occurred during the study have been authorized/acknowledged by the Study Director, assessed for impact, and documented in the study records. Only minor SOP deviations that did not impact the quality or integrity of the study occurred during the course of the study.

Appendix 2
Test Item Characterization

CORDENPHARMA

Certificate of Analysis – Revision 01

Product: MOD-201MS



Chemical Formula: $C_{44}H_{87}NO_5$
Molecular Weight: 710.18

Corden Pharma Sample Name: LP-04-285
Lot #: RL-100-211-1

Tests	Specifications	Results
Appearance	Clear oil, no foreign particulate matter visible	Clear oil, no foreign particulate matter visible
Color	Colorless to yellowish	Yellowish
Solubility (1w% in Ethanol)	Clear solution	Clear solution
Identity		
HPLC-CAD	Retention time of the main peak corresponds to reference	Retention time of the main peak corresponds to reference
IR-Spectrum	Comparable to reference	Comparable to reference
^{1}H-NMR	Conforms to reference	Conforms to reference
MS	M = 709.7 ± 1.0 m/z M + H$^+$ = 710.7 ± 1.0 m/z	Positive ionization mode: M + H$^+$ = 711.07 m/z
Purity		
Purity by CAD-HPLC (% Area)	Report result, target: ≥ 95% Report all imp. ≥ 0.05%	MOD-201MS purity = **95.72%** Impurities ≥ 0.05%: Unk. RRT: 0.66 0.20% Compound 7 RRT: 0.85 0.43% Unk. RRT: 0.93 0.07% Compound 11 RRT: 0.93 <0.05% Unk. RRT: 1.02 1.01% Unk. RRT: 1.03 0.21% Unk. RRT: 1.03 0.26% Unk. RRT: 1.04 0.29% Unk. RRT: 1.05 0.18% Unk. RRT: 1.05 0.26% Unk. RRT: 1.06 0.43% Unk. RRT: 1.06 0.12% Unk. RRT: 1.08 0.06% Unk. RRT: 1.08 0.15% Unk. RRT: 1.09 0.16% Unk. RRT: 1.10 0.07% Unk. RRT: 1.11 0.28%

CORDENPHARMA

		Unk. RRT: 1.15 0.05% Unk. RRT: 1.18 0.07%
Assay		
Assay (solvent free, anhydrous, inorganics free) (%w/w)**	Report result	**95.48 %**
Water content, KF (% w/w)	≤ 1.0%	0.2%
Peroxides	Report result	1.5 mmol/Kg*
Residual solvents, HS-GC-FID	Acetonitrile ≤ 410 ppm Cyclopentyl-methyl ether ≤ 150 ppm n-Hexane ≤ 290 ppm Dichloromethane ≤ 600 ppm Ethanol ≤ 5000 ppm Heptane ≤ 5000 ppm Methanol ≤ 3000 ppm Ethyl acetate ≤ 5000 ppm	Acetonitrile < 100 ppm Cyclopentyl-methyl ether < 100 ppm n-Hexane < 100 ppm Dichloromethane 400 ppm Ethanol < 100 ppm Heptane < 100 ppm Methanol < 100 ppm Ethyl acetate < 100 ppm
Elemental impurities, ICP	Cd ≤ 0.2 ppm Pb ≤ 0.5 ppm As ≤ 1.5 ppm Hg ≤ 0.3 ppm Co ≤ 0.5 ppm V ≤ 1 ppm Ni ≤ 2 ppm Li ≤ 25 ppm Sb ≤ 9 ppm Cu ≤ 30 ppm Mg report result Si report result	Cd < 0.1 ppm Pb 0.1 ppm As < 0.1 ppm Hg < 0.1 ppm Co < 0.1 ppm V < 0.1 ppm Ni 0.1 ppm Li < 0.1 ppm Sb < 0.1 ppm Cu 0.1 ppm Mg 0.5 ppm Si 21.8 ppm
Residue on ignition (%w/w) USP <281>	≤ 0.1 %	< 0.1%

* Note: Hydroperoxide content 1 mmol/Kg = 34 ppm
**Assay = 100% - HPLC impurities (%) – water (%) – inorganics (%)

Recommended Storage: -20 °C ± 5 °C, Protect from light, moisture and air. Store under inert gas.

Date: October 27, 2016

Signature: [redacted]

[redacted]

Appendix 3
Results for Individual Cultures

Table 1 Results for Individual Cultures - 4-Hour Treatment in the Absence of S9 (0S9)

Treatment	Conc. (μg/mL)	Total # of Cells Counted	Counts Per Total Cells Cells with X # of Nuclei			CBPI	Total No. of BC Examined	No. of MBC
			1	2	>2			
Ethanol	-	525	125	387	13	1.8	1000	3
		523	122	376	25	1.8	1000	1
SM-102	3.25	§	§	§	§	§	§	§
		§	§	§	§	§	§	§
	5.68	§	§	§	§	§	§	§
		§	§	§	§	§	§	§
	9.95	§	§	§	§	§	§	§
		§	§	§	§	§	§	§
	17.4	§	§	§	§	§	§	§
		§	§	§	§	§	§	§
	30.5	§	§	§	§	§	§	§
		§	§	§	§	§	§	§
	53.3	§	§	§	§	§	§	§
		§	§	§	§	§	§	§
	93.3	§	§	§	§	§	§	§
		§	§	§	§	§	§	§
	163	581	131	421	29	1.8	1000	5
		513	90	382	41	1.9	1000	5
	286	514	105	362	47	1.9	1000	4
		542	112	403	27	1.8	1000	7
	500[ppt]	512	117	373	22	1.8	1000	1
		538	138	379	21	1.8	1000	1
NOC	0.25	512	324	134	54	1.5	521	27
		515	331	138	46	1.4	961	38

CBPI = Cytokinesis-Block Proliferation Index.
BC = Binucleated cells.
MBC = Micronucleated binucleated cells.
NOC = Nocodazole
§ = Not assessed
ppt = Precipitate visible in the culture medium at the end of treatment

Table 2 Results for Individual Cultures - 4-Hour Treatment in the Presence of S9 (+S9)

Treatment	Conc. (µg/mL)	Total # of Cells Counted	Counts Per Total Cells Cells with X # of Nuclei 1	2	>2	CBPI	Total No. of BC Examined	No. of MBC
Ethanol	-	509	103	393	13	1.8	1000	4
		523	126	380	17	1.8	1000	3
SM-102	3.25	§	§	§	§	§	§	§
		§	§	§	§	§	§	§
	5.68	§	§	§	§	§	§	§
		§	§	§	§	§	§	§
	9.95	§	§	§	§	§	§	§
		§	§	§	§	§	§	§
	17.4	§	§	§	§	§	§	§
		§	§	§	§	§	§	§
	30.5	§	§	§	§	§	§	§
		§	§	§	§	§	§	§
	53.3	§	§	§	§	§	§	§
		§	§	§	§	§	§	§
	93.3	§	§	§	§	§	§	§
		§	§	§	§	§	§	§
	163	524	97	398	29	1.9	1000	7
		500	126	354	20	1.8	1000	3
	286	520	113	388	19	1.8	1000	5
		514	118	375	21	1.8	1000	7
	500	509	143	353	13	1.7	1000	1
		513	91	407	15	1.9	1000	5
CP	10	517	295	222	0	1.4	1000	19
		567	315	251	1	1.4	1000	25

CBPI = Cytokinesis-Block Proliferation Index.
BC = Binucleated cells.
MBC = Micronucleated binucleated cells.
CP = Cyclophosphamide
§ = Not assessed

Table 3 Results for Individual Cultures - 24-Hour Treatment in the Absence of S9 (0S9)

Treatment	Conc. (µg/mL)	Total # of Cells Counted	Counts Per Total Cells Cells with X # of Nuclei 1	2	>2	CBPI	Total No. of BC Examined	No. of MBC
Ethanol	-	514	201	293	20	1.6	1000	3
		520	175	336	9	1.7	1000	4
SM-102	3.25	§	§	§	§	§	§	§
		§	§	§	§	§	§	§
	5.68	§	§	§	§	§	§	§
		§	§	§	§	§	§	§
	9.95	§	§	§	§	§	§	§
		§	§	§	§	§	§	§
	17.4	§	§	§	§	§	§	§
		§	§	§	§	§	§	§
	30.5	§	§	§	§	§	§	§
		§	§	§	§	§	§	§
	53.3	§	§	§	§	§	§	§
		§	§	§	§	§	§	§
	93.3	§	§	§	§	§	§	§
		§	§	§	§	§	§	§
	163	506	152	337	17	1.7	1000	2
		529	173	342	14	1.7	1000	1
	286	518	203	296	19	1.6	1000	3
		536	170	351	15	1.7	1000	1
	500[ppt]	517	177	323	17	1.7	1000	2
		514	208	290	16	1.6	1000	3
MMC	0.10	516	204	310	2	1.6	1000	20
		505	162	340	3	1.7	1000	19

CBPI = Cytokinesis-Block Proliferation Index.
BC = Binucleated cells.
MBC = Micronucleated binucleated cells.
MMC = Mitomycin C
§ = Not assessed
ppt = Precipitate visible in the culture medium at the end of treatment

Appendix 4
Dose Formulation Analysis Report

FINAL REPORT

Study Phase: Analytical Chemistry

Test Facility Study No. 9601568

TEST FACILITY:
Charles River Laboratories Montreal ULC
(b) (6) [redacted]

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LIST OF APPENDICES

1. SUMMARY

Dose formulation samples have been analyzed by High Performance Liquid Chromatography (HPLC) for the determination of SM-102.

The dose formulations were within specification.

2. INTRODUCTION

This report describes the analytical evaluation of SM-102 in dose formulations (Ethanol) from Study No. 9601568.

For the work detailed in this report, the analytical phase experimental start date was 12 Sep 2016, and the analytical phase experimental completion date was 13 Sep 2016.

3. EXPERIMENTAL DESIGN

Analysis of dose formulations was carried out with regard to concentration.

During the study, duplicate samples were taken from each Test Item formulation (Dose Number 1 to 10). The samples were analyzed on the day of preparation.

4. MATERIALS AND METHODS

4.1. Materials

4.1.1. Reference Standard

Identification:	SM-102
Lot No.:	RL-100-211-1
Purity (Assay):	95.48% (95.3% used for calculations)
Retest Date:	27 Oct 2017
Storage Conditions:	Kept in a freezer set to maintain -20°C
Supplier:	Moderna Therapeutics, Inc.

4.1.2. Characterization of Reference Standard

The Supplier provided the documentation for the identity, strength, purity, composition, and stability for the reference standard. A copy of the supplied Certificate of Analysis or equivalent documentation are presented in Appendix 2.

4.1.3. Inventory and Disposition of Reference Standard

Records of the receipt, distribution, and storage of the reference standard was maintained. All unused Sponsor-supplied reference standard was returned to the Sponsor.

4.2. Methods

4.2.1. Analytical Procedure

The method of analysis is documented in analytical procedure AP.9601568.SL.01 (Appendix 1) and was previously validated under Study No. 1801841. Prior to study commencement, stability data were generated by the department of Analytical Chemistry, Charles River, CR MTL for up to 5 days, for formulations stored stored at ambient temperature and in a refrigerator set to maintain a temperature of 4°C, respectively, over the concentration range of 0.00900 - 60.0 mg/mL, under Study No. 1801841.

4.3. Computerized Systems

Critical computerized systems used in this study phase are listed below (see Text Table 1).

Text Table 1
Computerized Systems

System Name	Version No.	Description of Data Collected and/or Analyzed
Empower 3 (Waters Corporation)	Build 3471 SR1	Data acquisition for dose formulation analysis, including regression analysis and measurement of concentration and recovery of dose formulations using HPLC
Mesa Laboratories AmegaView CMS	v3.0 Build 1208.8	Continuous Monitoring System. Monitoring of standalone fridges, freezers, incubators, and selected laboratories to measure temperature, relative humidity, and CO2, as appropriate
Johnson Controls Metasys	MVE 5.4 (M5)	Building Automation System. Control of HVAC and other building systems, as well as temperature/humidity control and trending in selected laboratories and animal rooms

5. RESULTS AND DISCUSSIONS

The results of all samples were found to be within the acceptance criteria of ± 10% of their theoretical concentrations for stock solutions (dose number 10) and ± 15% of their theoretical concentrations for lower level solutions (dose number 1 to 9). Results are presented in Table 1.

All results presented in the tables of the report are calculated using non-rounded values as per the raw data rounding procedure and may not be exactly reproduced from the individual data presented.

6. CONCLUSION

The dose formulations were within specification.

7. REPORT APPROVAL

(b) (6)__________________ Date: 12 Jan 2017

Table 1 Study Samples–Concentration

Preparation Date	Dose Number	Theoretical Concentration		Measured Concentration (mg/mL)	Percent of Theoretical	Mean Measured Concentration (mg/mL)	Mean Percent of Theoretical
		µg/mL	mg/mL				
12 Sep 2016	1	(b) (4) [redacted]					
	2						
	3						
	4						
	5						
	6						
	7						
	8						
	9						
	10						

Appendix 1
Analytical Procedure

Analytical Procedure (AP.9601568.SL.01)

High Performance Liquid Chromatographic Method Using Ultraviolet/Visible Detection for the Determination of SM-102 in Dose Formulations

This analytical procedure can be used for the quantitation of SM-102 in dose formulation samples in ethanol over the concentration range of 0.00900 to 60.0 mg/mL.

Reference Standard and Vehicle

Reference Standard	SM-102
Lot number	RL-100-211-1
Purity	95.3% (to be used for calculations)
Vehicle:	Ethanol

For storage conditions for Reference Standard, refer to the corresponding log sheets or transfer sheets.

NOTES

- Modifications may be made to the conditions below in order to optimize the chromatography.
- All volumes and weights listed throughout this procedure may be scaled up or down as long as the final concentration remains the same.
- Any changes are to be noted in the raw data.
- Unless otherwise indicated, information relating to the time of mixing/stirring, temperature used in the preparation of solutions, diluents, mobile phases and vehicles will be considered non-critical.
- If a step is deemed critical, it will be noted within the procedure, and a positive entry will be made in the raw data. In addition, the use of sonication to aid in the dissolution of reagents or TI, will also be considered an appropriate method within the laboratory unless it is clearly indicated that sonication is not to be used.
- The analytical method was previously validated under study 1801841.

Chromatographic Conditions

(b) (4)

[redacted]

Reagents

Unless specified, reagents with grade (A.C.S., USP et al) or numerical purity will be used.

(b) (4) [redacted]

(b) (4) [redacted]

Analytical Procedure (AP.9601568.SL.01)

Preparation of Standards

(b) (4) [redacted]

Table 1: Preparation of standards

STD identification	Stock solution	Volume of stock solution (µL)	Final volume (mL)	Final concentration (µg/mL)
STD A	diluent	(b) (4) [redacted]		
STD B	STD STK A			
STD C	STD STK B			
STD D	STD STK A			
STD E	STD STK B			
STD F	STD STK A			
STD G	STD STK B			

Preparation of Blanks

(b) (4) [redacted]

Preparation of Spikes

(b) (4) [redacted]

Table 2: Dilution scheme for spiked samples

SPK ID	Replicate	Aliquot of SPK (µL)	Volume of acetonitrile to add (µL)	Flask volume (mL)	Final concentration (µg/mL)	Dilution factor
SPK A	n = 2	(b) (4) [redacted]				
SPK B	n = 2					

Analytical Procedure (AP.9601568.SL.01)

(b) (4) [redacted]

Table 3: Dilution scheme for samples

Dose number	Concentration		Initial Dilution			Final concentration (µg/mL)	Dilution factor
	(µg/mL)	(mg/mL)	Aliquot of sample (µL)	Volume of ACN to add (µL)	Flasks volume (mL)		
1	(b) (4) [redacted]						
2							
3							
4							
5							
6							
7							
8							
9							
10							

Injection Sequence

(b) (4) [redacted]

Analytical Procedure (AP.9601568.SL.01)

Calculations

System suitability

- Calculate relative standard deviation (%RSD) in response of CAL (n = 3) using the following equation:
  $\% RSD = (SD \div A) \times 100$
  SD – standard deviation in response
  A – average response
- Calculate system stability using the following equation:
  $(A_2 - A_1) \div A_1 \times 100$
  A_1 – average response (n = 3) of CAL at the beginning of the run.
  A_2 – average response (n = 3) of CAL at the end of the run.

Standard curve

- Perform the least squares fit regression of peak area versus concentration (type of curve fit: **Linear**; weighting factor: **$1/x^2$**)

Calculation of concentrations

- Using Empower 3 Custom Field, calculate concentrations and accuracies of spikes and study samples.

Acceptance Criteria

Unless specified in the study plan or below, refer to SOP CAD-002 and SOP CAD-003 for acceptance criteria.

Study Samples

(b) (4)

AP Version Control

Initial version.

Verified by (b) (6) Date 09 Sep 2016

Approved by (b) (6) Date 09Sep 2016.

Individual Scien

Authorized by (b) (6) Date 09 Sep 2016

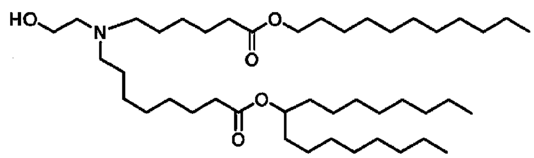
Scientific Director

Appendix 2
Certificate of Analysis

CORDENPHARMA

Certificate of Analysis – Revision 01

Product: MOD-201MS



Chemical Formula: $C_{44}H_{87}NO_5$
Molecular Weight: 710.18

Corden Pharma Sample Name: LP-04-285
Lot #: RL-100-211-1

Tests	Specifications	Results
Appearance	Clear oil, no foreign particulate matter visible	Clear oil, no foreign particulate matter visible
Color	Colorless to yellowish	Yellowish
Solubility (1w% in Ethanol)	Clear solution	Clear solution
Identity		
HPLC-CAD	Retention time of the main peak corresponds to reference	Retention time of the main peak corresponds to reference
IR-Spectrum	Comparable to reference	Comparable to reference
^{1}H-NMR	Conforms to reference	Conforms to reference
MS	M = 709.7 ± 1.0 m/z M + H$^+$ = 710.7 ± 1.0 m/z	Positive ionization mode: M + H$^+$ = 711.07 m/z
Purity		
Purity by CAD-HPLC (% Area)	Report result, target: ≥ 95% Report all imp. ≥ 0.05%	MOD-201MS purity = **95.72%** Impurities ≥ 0.05%: Unk. RRT: 0.66 0.20% Compound 7 RRT: 0.85 0.43% Unk. RRT: 0.93 0.07% Compound 11 RRT: 0.93 <0.05% Unk. RRT: 1.02 1.01% Unk. RRT: 1.03 0.21% Unk. RRT: 1.03 0.26% Unk. RRT: 1.04 0.29% Unk. RRT: 1.05 0.18% Unk. RRT: 1.05 0.26% Unk. RRT: 1.06 0.43% Unk. RRT: 1.06 0.12% Unk. RRT: 1.08 0.06% Unk. RRT: 1.08 0.15% Unk. RRT: 1.09 0.16% Unk. RRT: 1.10 0.07% Unk. RRT: 1.11 0.28%

CORDENPHARMA

		Unk. RRT: 1.15 0.05% Unk. RRT: 1.18 0.07%
Assay		
Assay (solvent free, anhydrous, inorganics free) (%w/w)**	Report result	**95.48 %**
Water content, KF (% w/w)	≤ 1.0%	0.2%
Peroxides	Report result	1.5 mmol/Kg*
Residual solvents, HS-GC-FID	Acetonitrile ≤ 410 ppm Cyclopentyl-methyl ether ≤ 150 ppm n-Hexane ≤ 290 ppm Dichloromethane ≤ 600 ppm Ethanol ≤ 5000 ppm Heptane ≤ 5000 ppm Methanol ≤ 3000 ppm Ethyl acetate ≤ 5000 ppm	Acetonitrile < 100 ppm Cyclopentyl-methyl ether < 100 ppm n-Hexane < 100 ppm Dichloromethane 400 ppm Ethanol < 100 ppm Heptane < 100 ppm Methanol < 100 ppm Ethyl acetate < 100 ppm
Elemental impurities, ICP	Cd ≤ 0.2 ppm Pb ≤ 0.5 ppm As ≤ 1.5 ppm Hg ≤ 0.3 ppm Co ≤ 0.5 ppm V ≤ 1 ppm Ni ≤ 2 ppm Li ≤ 25 ppm Sb ≤ 9 ppm Cu ≤ 30 ppm Mg report result Si report result	Cd < 0.1 ppm Pb 0.1 ppm As < 0.1 ppm Hg < 0.1 ppm Co < 0.1 ppm V < 0.1 ppm Ni 0.1 ppm Li < 0.1 ppm Sb < 0.1 ppm Cu 0.1 ppm Mg 0.5 ppm Si 21.8 ppm
Residue on ignition (%w/w) USP <281>	≤ 0.1 %	< 0.1%

* Note: Hydroperoxide content 1 mmol/Kg = 34 ppm
**Assay = 100% - HPLC impurities (%) – water (%) – inorganics (%)

Recommended Storage: -20 °C ± 5 °C, Protect from light, moisture and air. Store under inert gas.

Date: October 27, 2016

Signature: (b) (6)

[redacted]

Appendix 5
Historical Control Results

These QA audited results were collected from non-GLP and GLP compliant *in vitro* micronucleus studies performed from February 2009 to February 2015.

Negative Controls

Regime	% MBC	SD	Mean incidence of MBC per 1000 binucleate cells	Upper 95% limit[1] (% MBC)	Upper 99% limit[1] (% MBC)	No. of Treatments
4-hour treatment without S9	0.42	0.19	4.2	1.024	1.259	12
4-hour treatment with S9	0.68	0.47	6.8	1.442	1.713	36
long treatment without S9	0.52	0.45	5.2	1.167	1.415	32

MBC = Micronucleated binucleate cells; SD = Standard deviation; 1 = Poisson-based control limits as per the Report on Statistical Issues Related to OECD Test Guidelines (TGs) on Genotoxicity, No. 198 (2014) published by the OECD, limits were subsequently converted into %.

Positive Controls

Positive Control and Regime	% MBC	SD	No. of Treatments
Mitomycin C (4-hour without S9)	13.04	5.98	12
Cyclophosphamide (4-hour with S9)	6.58	3.28	31
Mitomycin C (long exposure without S9)	14.56	5.64	29
Colcemid (long exposure without S9)	4.22	6.6	11

MBC = Micronucleated binucleate cells; SD = Standard deviation